(Rule 201(b)(1)(i)), would have the option of obtaining proposed consolidated market data from one or more competing consolidators (rather than from a single plan processor as is currently the case), to aggregate consolidated market data itself, or some combination of the two.[521]

The Commission also proposed certain conforming amendments in Rule 201 to harmonize that rule with the Proposing Release. Currently, 17 CFR 242.201(a) (Rule 201(a)) defines "listing market" by reference to the listing market as defined in the effective transaction reporting plan for the covered security. Since primary listing exchanges will be required to collect and calculate regulatory data, the Commission proposed to introduce a definition of "primary listing exchange" in Rule 600(b)(68) to provide greater clarity with respect to the responsibilities regarding regulatory data. Specifically, under proposed Rule 600(b)(68), primary listing exchange would be defined as, for each NMS stock, the national securities exchange identified as the primary listing exchange in the effective national market system plan or plans required under Rule 603(b).[522]

The Commission believes that it is appropriate for the effective national market system plan(s) to determine which exchange is the primary listing exchange for each NMS stock and that the definition would ensure that primary listing exchanges are clearly identified. The Commission also believes that the definition of listing market in Rule 201(a)(3) should be amended so that it cross-references this proposed definition of primary listing exchange to facilitate the consistent identification of primary listing exchanges across Regulation SHO and

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521 For example, a listing market could self-aggregate for its own listings and obtain consolidated data from a competing consolidator for stocks listed elsewhere. The rules that the Commission is adopting do not require a listing market to purchase consolidated data from a competing consolidator.

522 See infra Section III.E.2(j) (discussing the requirement that the effective national market system plan(s) be amended to include a list of the primary listing exchange for each NMS stock).

Regulation NMS and to avoid potentially duplicative or confusing definitions in the Commission's rules.

2. Final Rule and Response to Comments

The Commission received one letter supporting the amendments to Regulation SHO.[523] For the same reasons discussed above with regard to how the proposed amendments would facilitate the decentralized consolidation model, the Commission is adopting the amendments to Regulation SHO as proposed.

J. Definition of "Administrative Data" under Rule 600(b)(2)

1. Proposal

The Commission proposed defining "administrative data" as administrative, control, and other technical messages made available by national securities exchanges and national securities associations pursuant to the effective national market system plan or plans required under Section 242.603(b) or the technical specifications thereto as of [date of Commission approval of this proposal]. Administrative data would be a component of the definition of "consolidated market data," which permits additional administrative data elements to be added pursuant to amendments to the effective national market system plan(s). Examples of administrative messages include market center and issue symbol identifiers, and examples of control messages include messages regarding the beginning and end of trading sessions. As the Commission stated in the proposing release, the proposed definition was "intended to capture administrative information that is currently provided in SIP data."[524]

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523 See Data Boiler Letter I at 27.

524 See Proposing Release, 85 FR at 16763.

2. Final Rule and Response to Comments

The Commission received one comment supporting the proposed definition of administrative data,[525] and is adopting the definition as proposed. The Commission continues to believe that including administrative messages in consolidated market data will facilitate market participants' efficient and accurate use of consolidated market data. Further, the Commission believes that this information is useful to market participants and should continue to be widely available. The Commission believes that SROs would be well-situated to provide administrative data messages, which relate to SRO-specific details such as the market-center identifiers or the beginning and ending of trading sessions, because SROs have direct and immediate access to this information and could efficiently integrate it into the data feeds that they will utilize to make available the data necessary for competing consolidators and self-aggregators to generate core and regulatory data.

K. Definition of "Self-Regulatory Organization-Specific Program Data" under Rule 600(b)(85)

1. Proposal

The Commission proposed to define exchange-specific program data as: (i) information related to retail liquidity programs specified by the rules of national securities exchanges and disseminated pursuant to the effective national market system plan or plans required under

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525 This commenter stated that the current administrative data "provides additional context for market participants to understand, and efficiently and accurately use, the proposed core and regulatory data to support their trading activities." Data Boiler Letter I at 35. The commenter further added that "there can be streamlining opportunity for [competing consolidators] to eliminate any repetitive information during distribution and recipients should have a choice to opt-out." Id. As stated above, the decentralized consolidation model permits competing consolidators to offer different products to market data end users, allowing end users to decide which data feeds to purchase and utilize.

Section 242.603(b) as of [date of Commission approval of this proposal] and (ii) other exchange-specific information with respect to quotations for or transactions in NMS stocks as specified by the effective national market system plan or plans required under Section 242.603(b).

The Commission stated that existing retail liquidity programs, which offer opportunities for retail orders to receive price improvement, and, in certain cases, other exchange-specific program information should continue to be included in proposed consolidated market data. If (i) an exchange(s) develops new program(s) in the future, and (ii) the broad dissemination of information about such programs as part of consolidated market data would facilitate participation in such programs, an amendment to the effective national market system plan(s) could be filed with the Commission under Rule 608 of Regulation NMS to include such information in consolidated market data.

2. Final Rule and Response to Comments

The Commission received comments regarding the definition of exchange-specific program data. One commenter supported the inclusion of exchange-specific program data in consolidated market data, stating that this data, which is already carried by the SIPs, is highly relevant and important to all types of market participants.[526] Another commenter agreed with the inclusion of information related to existing retail liquidity programs but stated that there should be a "procedural mechanism to review if there might be other new exchange-specific program information to be included in the future."[527]

Some commenters objected to the proposed definition of exchange-specific program data. One commenter stated that the proposal would require changes to exchange-specific programs to

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[526] See IEX Letter at 7.

[527] Data Boiler Letter I at 35–36.

become effective through an effective national market system plan amendment even though exchanges are currently free to propose such programs through the SRO rulemaking process provided in Section 19(b) of the Exchange Act and that requiring such changes to be duplicatively filed as proposed plan amendments would serve no policy or regulatory purpose and would improperly give competing exchanges (as members of the plans' Operating Committee) a vote in whether or not an exchange may change its programs.[528] Another commenter characterized exchange-specific program information as "an essentially unknown category of information that may or may not be useful to particular categories of investors" and stated that the Commission has determined that "virtually all categories of information – even indeterminate ones – constitute core data."[529]

The Commission is adopting the definition of exchange-specific program data largely as proposed, but is modifying the definition to "self-regulatory organization-specific program data" so that it extends to national securities associations in addition to all national securities exchanges.[530] The Commission believes that information related to any program developed by a national securities association in the future should also be able to be included in consolidated market data if specified by the effective national market system plan(s).[531] Information related to

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528 See NYSE Letter II at 28.

529 Nasdaq Letter IV at 33.

530 The Commission is also modifying proposed Rule 600(b)(85)(ii) of Regulation NMS so that it refers to "[o]ther self-regulatory organization-specific information with respect to quotations for or transactions in NMS stocks . . ." rather than "other exchange-specific information with respect to quotations for or transactions in NMS stocks . . .". See infra Section VIII.

531 See also supra Section II.F.2(c) (explaining that the Commission is adopting a modified definition of depth of book data that includes liquidity at depth of book price levels that may be available on FINRA's ADF or other facilities of a national securities association in the future).

retail liquidity programs is already in the SIP feeds today, and the Commission agrees that this information is relevant to market participants who wish to submit orders to, or otherwise participate in, such programs and should therefore be included in consolidated market data. To the extent other exchange-specific or national securities association-specific programs may be developed in the future, the Commission believes such information would be similarly relevant to market participants who wish to engage with such programs.

The Commission also agrees that a procedural mechanism to modulate the inclusion in consolidated market data of information related to future SRO-specific programs is needed. The Commission believes that requiring such data to be included by amendment to the effective national market systems plan(s), as proposed, is the appropriate mechanism because, as explained above, it allows for the inclusion of additional SRO-specific program information data elements that may emerge periodically through the approval of new SRO rules.[532]

The Commission disagrees with the comment that this process would be "duplicative" of the Section 19(b) process or give an exchange's competitors a vote in whether an exchange may change its programs. The Section 19(b) process for approval of an SRO's rules and a plan amendment serve two distinct purposes. An individual SRO could develop a new program on its own initiative pursuant to the Section 19(b) process. The SRO could disseminate information related to any such program to market participants on a proprietary basis only. On the other hand, a plan amendment would only be required in order to include this information in consolidated market data if an SRO decides to pursue the option and the Operating Committee agrees it is appropriate to provide it to market participants under the national market system rules.

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532 See supra Section II.B.2.

Finally, the Commission disagrees that exchange-specific program data is an "essentially unknown category of information" that may or may not be useful to market participants. The rules would allow the effective national market system plan(s) to add other SRO-specific information if the Operating Committee determines that the information would be useful to market participants. The Commission believes that allowing the Operating Committee(s) some flexibility to add additional SRO-specific information would be in the interest of investors and would strengthen the national market system. The Equity Data Plans have utilized such a mechanism in the past.

III. Enhancements to the Provision of Consolidated Market Data

A. Introduction

The Commission is adopting a decentralized consolidation model in which competing consolidators, rather than the exclusive SIPs, will collect, consolidate, and disseminate consolidated market data. This new model will address the geographic, aggregation, and transmission latencies that characterize the existing centralized consolidation model,[533] which has relied upon the exclusive SIP for each NMS stock to centrally collect, consolidate, and

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533 See Proposing Release, 85 FR at 16765–66 (describing the latencies that exist in the current centralized consolidation model). The existing centralized consolidation model system suffers from three sources of latencies: (a) geographic latency, (b) aggregation latency, and (c) transmission latency. Geographic latency is typically the most significant component of the latencies that the exclusive SIPs experience compared to the proprietary data feeds. Geographic latency, as used herein, refers to the time it takes for data to travel from one physical location to another, which must also take into account that data does not always travel between two locations in a straight line. Aggregation or consolidation latency, as used herein, refers to the amount of time an exclusive SIP takes to aggregate the multiple sources of SRO market data into SIP data and includes calculation of the NBBO. Transmission latency, as used herein, refers to the time interval between when data is sent (e.g., from an exchange) and when it is received (e.g., at an exclusive SIP and/or at the data center of the subscriber), and the transmission latency between two fixed points is determined by the transmission communications technology through which the data is conveyed.

disseminate SIP data from its location, regardless of the location of other exchanges, FINRA, or subscribers. The Commission believes this new model will foster a competitive environment for the dissemination of consolidated market data and will modernize the underlying architecture of the national market system.

The centralized consolidation model has largely remained unchanged despite significant market developments since it was developed in the 1970s. Today, the exclusive SIPs are located in disparate locations far from each other and from end users. Each exclusive SIP must collect data from geographically dispersed SRO data centers, consolidate that data, and then disseminate that data as SIP data from the exclusive SIP's location to end-users, which are often in other locations. The need for market information to travel back and forth across the "New Jersey triangle"[534] prior to reaching subscribers creates significant geographic and other latencies.[535] This structure, as well as the limited data content available in SIP data, has led many market participants to relegate SIP data to backup data.[536]

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534 The CTA/CQ SIP is located in Mahwah, NJ, and the UTP SIP is located in Carteret, NJ. Other exchanges and broker-dealers are located in Secaucus, NJ. These three main data center locations are typically referred to as the "New Jersey Triangle."

535 See supra note 533.

536 See, e.g., SIFMA Letter at 3 ("Even if a broker-dealer elects to consolidate market data through proprietary feeds, it must also purchase the core data from the SIPs for a number of reasons, such as to comply with the Vendor Display Rule, receive regulatory messages like trading halts and have a backup source of data in case an exchange experiences issues with its proprietary feeds."). But see BestEx Research Letter at 2 ("Despite the claims of many market participants, the SIP is a critical component of the US equity market structure and is widely used by institutional broker-dealers."). The commenter, however, also stated that "the reforms to the SIP proposed by the Commission will make it even more robust and useful." Id. at 1. While SIP data is widely used, market participants, as well as the commenter, acknowledge that the decentralized consolidation

The national market system for the collection, consolidation, and dissemination of SIP data was established to be the heart of the national market system and is designed to provide broad public access to a consolidated, real-time stream of market information.[537] Investors' need for real time information has been recognized since the adoption of the 1975 Amendments and is reflected in Section 11A of the Exchange Act.[538] In the context of market data, the Commission has said that "real time" means that "there is very little delay between the time a quotation is made or a transaction is effected and the time that this information is made available to investors and others who use the information."[539] Some market participants believe that the latencies inherent in the centralized consolidation model have affected the ability of brokers to trade competitively and to provide best execution to customer orders, especially when compared to proprietary data products that are not encumbered by centralized consolidation.[540]

Significant technological changes have occurred since the 1970s and the passage of the 1975 Amendments. Electronic trading has all but supplanted manual trading, and electronic trading systems can handle and process data at speeds unheard of when the national market system was established. While the Equity Data Plans have made various investments and

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model will modernize the national market system and make it more useful for today's trading.

537 See Market Information Concept Release, supra note 22, at 70614.

538 Section 11A(c)(1)(B) of the Exchange Act states that the Commission should assure, among other things, the "prompt" collection, processing, distribution, and publication of information. Further, the Senate Report for the enactment of Section 11A stated that "it is critical for those who trade to have access to accurate, up-to-the-second information." S. REP. NO. 94-75 at 8 (1975) ("Senate Report").

539 Market Information Concept Release, supra note 22, at 70614.

540 One commenter stated that "[c]urrent market structure allows investors' order [sic] to be traded at stale prices." Better Markets Letter at 6. See also Capital Group Letter at 2, 4; Clearpool Letter at 2; DOJ Letter at 2, 4; Fidelity Letter at 2; MFA Letter at 2; State Street Letter at 2, 3; T. Rowe Price Letter at 1–2; Virtu Letter at 2, 5.

systems upgrades over time, they have not kept up with the demands of all market participants. The concurrent existence of the centralized consolidation model for SIP data and the decentralized consolidation model for proprietary data has resulted in a two-tiered market in which certain market participants that can afford and choose to pay for proprietary data feeds receive content-rich data faster than those who do not purchase these feeds, including market participants who may face higher barriers to entry from data and other exchange fees. Market participants that do not receive proprietary DOB feeds may be affected in their efforts to seek best execution and otherwise effectively compete with market participants that receive proprietary DOB data feeds because they do not obtain access to the additional content and may be receiving data in a slower manner.

Therefore, the Commission believes that the national market system must be modernized to allow "new data processing and communications systems [to] create the opportunity for more efficient and effective market operations."[541] The centralized consolidation model no longer meets market participants' need for real-time consolidated market data.[542] The purpose of the decentralized consolidation model is to modernize the infrastructure of the national market system by eliminating the outdated centralized architecture for data consolidation and dissemination.

Under the current model, each exclusive SIP must collect data for specific NMS stocks from geographically dispersed SRO data centers, consolidate the data, and then disseminate it

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541 Section 11A(a)(1)(B) of the Exchange Act.

542 See supra Section II.C.2(a) (discussing the indirect benefits to market participants whose executing broker-dealers will receive expanded data content from competing consolidators); infra Section III.B.5 (discussing indirect benefits to investors from enhancements to trading by their broker-dealers resulting from reductions in latency, the expanded data content and the competitive environment fostered by the decentralized consolidation model).